

ASP.NET Core Web API Microservices

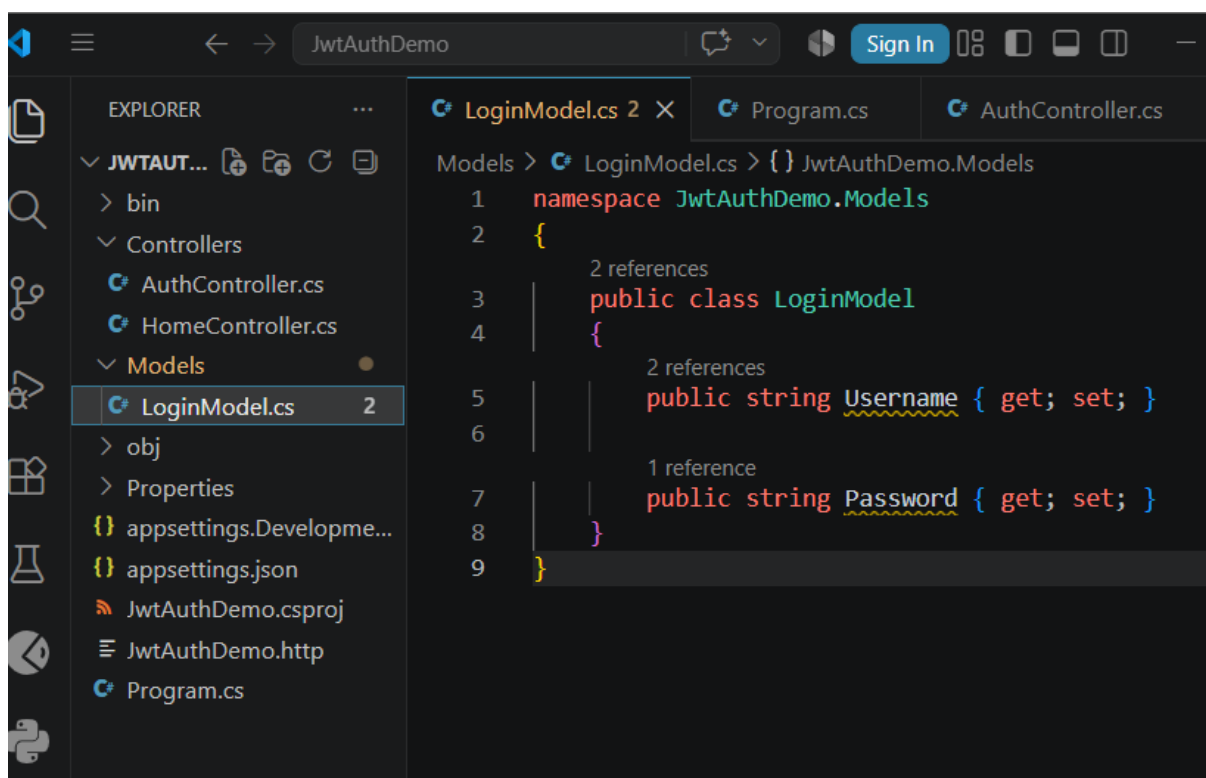
Question 1: Implement JWT Authentication in ASP.NET Core Web API

Steps

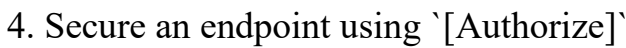
1: Create a new ASP.NET Core Web API project

```
PS C:\Users\laksh\FirstWebAPI> cd ..  
PS C:\Users\laksh> dotnet new webapi -n JwtAuthDemo  
The template "ASP.NET Core Web API" was created successfully.
```

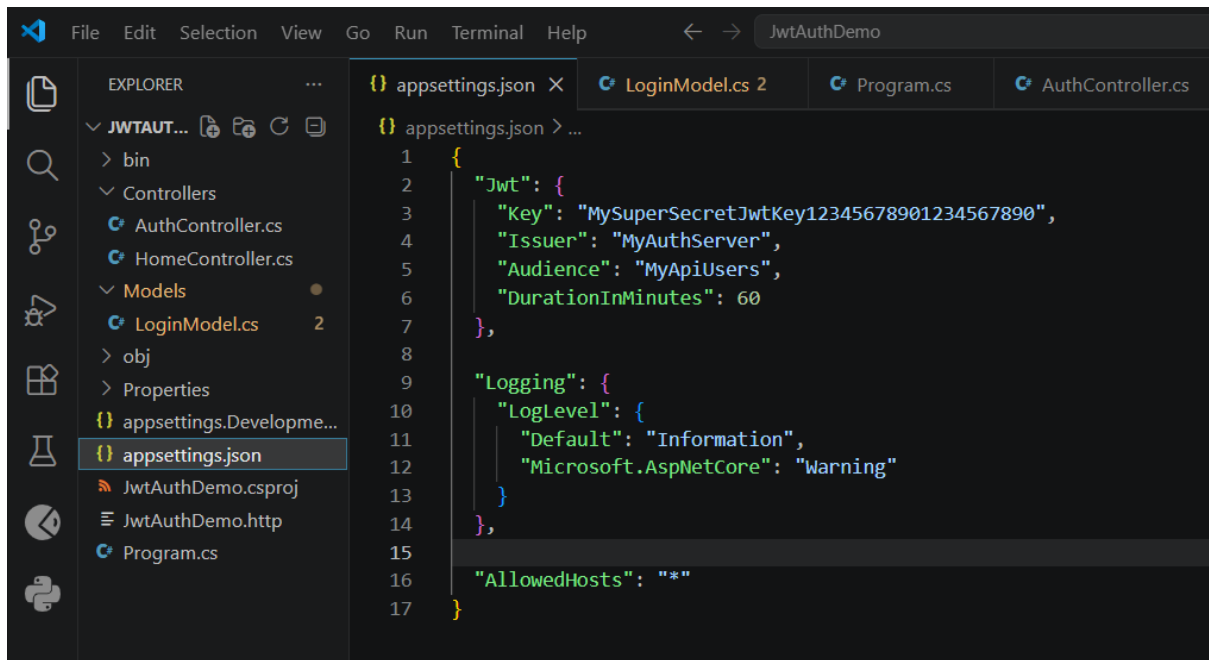
2. Add a 'User' model and a login endpoint



3. Generate a JWT token upon successful login.

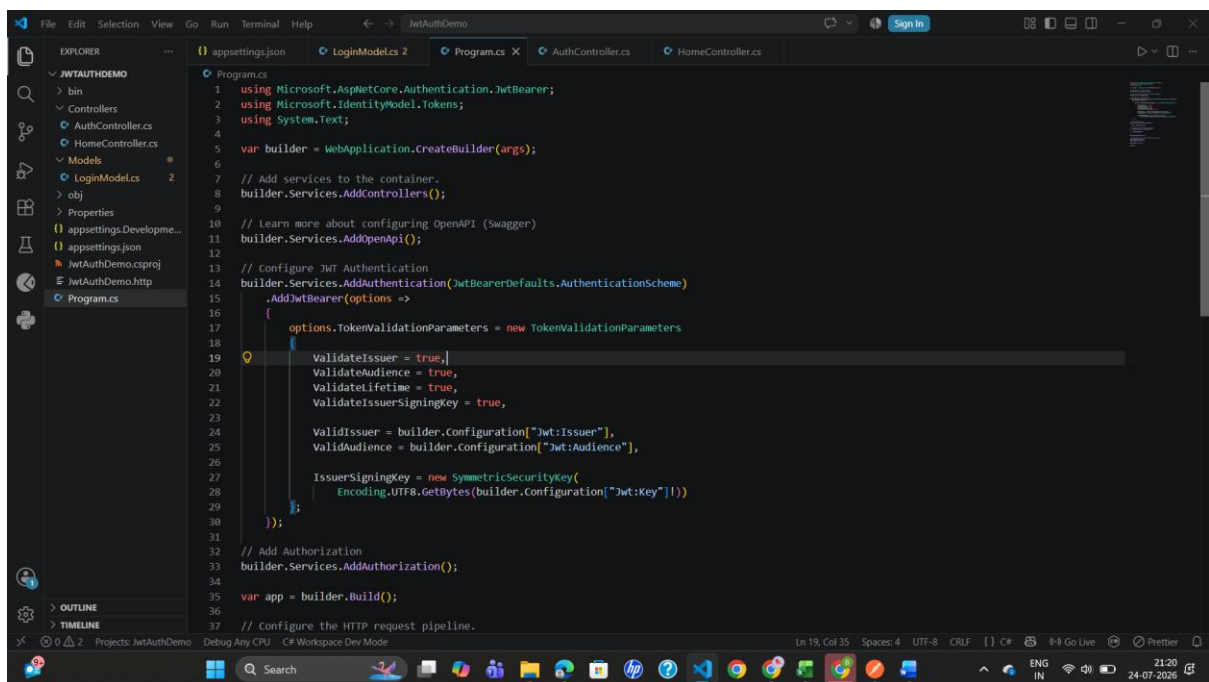


Appsetting.json:



```
1 {
2   "Jwt": {
3     "Key": "MySuperSecretJwtKey12345678901234567890",
4     "Issuer": "MyAuthServer",
5     "Audience": "MyApiUsers",
6     "DurationInMinutes": 60
7   },
8
9   "Logging": {
10    "LogLevel": {
11      "Default": "Information",
12      "Microsoft.AspNetCore": "Warning"
13    }
14  },
15
16  "AllowedHosts": "*"
17 }
```

Program.cs:



```
1 using Microsoft.AspNetCore.Authentication.JwtBearer;
2 using Microsoft.IdentityModel.Tokens;
3 using System.Text;
4
5 var builder = WebApplication.CreateBuilder(args);
6
7 // Add services to the container.
8 builder.Services.AddControllers();
9
10 // Learn more about configuring OpenAPI (Swagger)
11 builder.Services.AddOpenApi();
12
13 // Configure JWT Authentication
14 builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme)
15     .AddJwtBearer(options =>
16     {
17         options.TokenValidationParameters = new TokenValidationParameters
18         {
19             ValidateIssuer = true,
20             ValidateAudience = true,
21             ValidateLifetime = true,
22             ValidateIssuerSigningKey = true,
23
24             ValidIssuer = builder.Configuration["Jwt:Issuer"],
25             ValidAudience = builder.Configuration["Jwt:Audience"],
26
27             IssuerSigningKey = new SymmetricSecurityKey(
28                 Encoding.UTF8.GetBytes(builder.Configuration["Jwt:Key"]!))
29         };
30     });
31
32 // Add Authorization
33 builder.Services.AddAuthorization();
34
35 var app = builder.Build();
36
37 // Configure the HTTP request pipeline.
```

Authcontrol.cs:

